PRANAY DAS

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PROFESSIONAL EXPERIENCE

Eli Lilly and Company **October 2024 - Present**
Software Developer *Remote*
- • Developed a concurrent slot reservation system using idempotent requests and version-based optimistic concurrency control, preventing double bookings of the same time slot for clinical operators.
- Implemented correlation IDs to trace events across services and added role- and group-based access controls using Python and FastAPI.
- Developed a Python/FastAPI verification and routing service within an agentic adverse event processing system, validating classified cases, routing mismatches for human review, and sending verified cases through Temporal workflows to MuleSoft/Veeva, enabling ~70% of cases to bypass manual classification review.
- Built a component to verify agent-generated citations against retrieved evidence from a collection of about 50K labeled documents.

Google Summer of Code **May 2024 - August 2024**
Open Source Software Developer *Remote*
- Built and published scorecode, a Python package on PyPI that retrieves software supply-chain security metrics from the OpenSSF Scorecard API.
- Developed reusable Django models and a ScanCode.io integration to store Scorecard results and include them in CycloneDX SBOM exports.
- Collaborated with open-source maintainers through iterative code reviews, refining API design, class naming conventions, documentation, and code quality to meet project contribution standards.
- Implemented automated pytest suites covering API response validation, data integrity, and edge cases across public, private, and nonexistent repositories.

Instoried **August 2019 - May 2023**
Software Developer (Founding Engineer) *Bengaluru, Karnataka, India*
- Built the backend platform from scratch using Python, FastAPI, and PostgreSQL, developing REST APIs and recommendation engine integrations for a SaaS product that scaled to 100K+ users.
- Led a team of 5-6 backend, frontend, and DevOps engineers, taking part in technical design, code reviews, hiring, and mentoring alongside the CTO.
- Built a centralized OAuth 2.0 authorization service and developer portal that allowed customers to use the recommendation APIs directly, extending access beyond the web application.
- Developed a Git-like text versioning system using snapshots and incremental deltas, reducing database storage overhead, read/write latency, and payload sizes by transmitting only modified text chunks.
- Used Redis caching with content-based keys, TTLs, and LFU eviction to reuse recommendation results and reduce repeated processing.
- Developed an incremental ETL pipeline using checkpoint-based PostgreSQL queries to synchronize user onboarding and campaign data with a marketing analytics dashboard, reducing load on the primary database and enabling near-real-time sales insights.
- Configured Kubernetes horizontal pod autoscaling and load balancing to handle campaign-driven traffic spikes, alongside Jenkins CI/CD pipelines and pytest-based automated testing.

KaptureCRM **January 2019 - May 2019**
Software Developer *Bengaluru, Karnataka, India*
- Designed and developed a bidirectional data synchronization platform using a canonical data model to integrate Shopify, HubSpot, NetSuite, Snowflake, and SQL databases, enabling seamless CRM integration for 50+ onboarded clients.
- Implemented a hybrid synchronization pipeline using webhooks, WAL-based CDC, and offset-based polling workers, achieving near-real-time data propagation with eventual consistency across connected systems.

Aptus Data Labs **June 2017 - December 2018**
Software Developer *Bengaluru, KA, India*
- Developed 5 REST API connectors for external applications and databases as part of a data integration platform supporting different SQL/NoSQL data sources, enabling configurable cross-source data migration.
- Designed a unified data schema that eliminated source-to-destination-specific transformation logic, enabling reusable data mappings and simplifying integration across heterogeneous systems.
- Built Java-based Azure and JSON connector plugins for Yellowfin BI, expanding data connectivity for business intelligence reporting and visualization.

SKILLS

Programming Languages: Python, C/C++, Java
Web Frameworks & APIs: FastAPI, Flask, Django, Node.js, REST APIs, gRPC, GraphQL, OpenAPI, Swagger, pytest
Cloud & DevOps: AWS, S3, Lambda, Docker, Kubernetes, Git, CI/CD, Terraform, Jenkins, Google Cloud Platform, BigQuery
Databases: PostgreSQL, Supabase, MySQL, MongoDB, Redis, Cassandra, Elasticsearch, DynamoDB
Architecture & Infrastructure: Temporal, Apache Kafka, Modular Monolith, Microservices

PROJECTS & OUTSIDE EXPERIENCE

Compliance Checker
Freelance Software Developer (Novo Nordisk) - Pharmaceuticals/Healthcare *India*
- Designed and developed the complete backend using Python and FastAPI to extract information from clinical PDFs, check documents against client-defined compliance rules, flag violations, and apply corrections.
- Replaced an EC2-based PDF processing service with parallel AWS Lambda functions, splitting 400-page documents into smaller tasks of 20–30 pages that could be processed in 6–10 seconds each.
- Built a reliable processing pipeline using DynamoDB to combine extraction results, with automatic retries and dead-letter queues to handle failed tasks and flag unresolved issues for developer review.

KYC Portal
Freelance Software Developer (Natixis Investments) - FinTech *India*
- Designed and developed a KYC/AML screening platform using Python, FastAPI, and PostgreSQL, replacing spreadsheet-based workflows and reducing screening and review turnaround from approximately one week to 8 hours.
- Built a Kafka-based pipeline to collect financial data from Moody's, Refinitiv World-Check, and internal banking APIs in parallel, combining results into a common data format for analysts to review.
- Implemented real-time screening updates using Server-Sent Events (SSE) and SSO-based access, allowing analysts to track progress, review reports, add comments, and collaborate across teams.

EDUCATION

Purdue University - West Lafayette **June 2023 - May 2025**
Master's, Computer Science

Sambalpur University **August 2013 - April 2017**
Bachelor's, Computer Science